Conner B. Draper Seokkwon Yoon

Math 402/436

Setup

Goal. Predict next-day stock return $r_{i,t+1}$ from a signal $z_{i,t}$ (cross-sectional z-score).

Standard model (Grinold & Kahn, 1999):

$$\mathbb{E}[r_{i,t+1} \mid z_{i,t}] = \mathrm{IC} \cdot \sigma_{i,t} \cdot z_{i,t}$$

$\mathrm{IC} =$ correlation between signal and returns. Think of it as *scaled confidence*: $\mathrm{IC} \approx 0.05$ means the signal is right ∼52% of the time.

Our extension. Condition IC on z and let it drift over time:

$$\mathbb{E}[r_{i,t+1} \mid z_{i,t}] = \mathrm{IC}_t(z_{i,t}) \cdot \sigma_{i,t} \cdot z_{i,t}$$

We estimate $\mathrm{IC}_t(\cdot)$ with kernel regression on a rolling 504-day window.

Motivation. The assumption that IC$=+0.05$ for all signals, across all time is an gross oversimplification. IC is likely a non-linear function of z that decays to 0 time.

Data

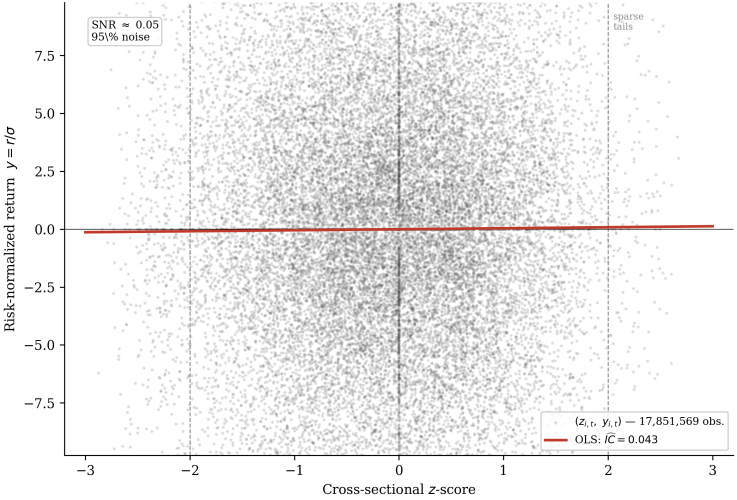
Barra U.S. equity model, 1995–2025

$\sim 3{,}000$ stocks/day

8 signals:

- Style / industry / idiosyncratic momentum
- Style / industry / idiosyncratic reversal
- Idiosyncratic volatility
- Betting-against-beta

We regress on $y_{i,t} = r_{i,t+1}/\sigma_{i,t}$ so OLS slope = IC directly.



Style momentum signal pooled across all stocks and dates (1995–2025).

Red line: OLS $\widehat{\mathrm{IC}} \approx 0.05$ — nearly flat against the noise.

Kernel Ridge Regression (KRR)

Mathematical Core. Learn $f : \mathbb{R} \to \mathbb{R}$ via regularized least squares in an RKHS:

$$\min_{f \in \mathcal{H}} \sum_{i=1}^{n} (f(z_i) - y_i)^2 + \alpha \|f\|_{\mathcal{H}}^2$$

Update Step (Daily Training).

- **Buffer:** Maintain a rolling 504-day window of cross-sectional (z, y) pairs.
- **Fit:** Heavy regularization $\alpha \approx 19$ (Noise-to-Signal ratio) forces high-bias/low-variance predictions to suppress noise.

Predict Step (Inference).

- Input today's z-scores. Because $y_i \approx \mathrm{IC} \cdot z_i$, the model outputs $\alpha_z = \widehat{\mathrm{IC}}(z) \cdot z$ directly.
- Output is unconstrained in sign, natively capturing negative IC.
- Uncertainty (variance) is treated as a static heuristic based on α.

Gaussian Process Regression (GPR)

Mathematical Core. Bayesian prior over functions $f \sim \mathcal{GP}(0, k)$. Kernel: $k(z, z') = \text{Constant} \times \text{RBF} + \text{WhiteKernel}$.

Update Step.

- Uses the exact same 504-day rolling buffer and 300-point recent subsample as KRR.
- **Fit:** WhiteKernel noise is *fixed* at 0.95 (representing 95% noise). This forces the exact posterior to ignore daily volatility and focus on smooth, persistent trends.

Predict Step.

- Output yields expected return $\mu_*(z)$ AND predictive variance $\sigma_*^2(z)$.
- **Key advantage:** Provides dynamic, point-wise uncertainty useful for downstream portfolio sizing (unlike KRR's flat variance).

Portfolio Construction

Alpha forecast:

$$\hat{\alpha}_{i,t} = \widehat{\mathrm{IC}}_t(z_{i,t}) \cdot \sigma_{i,t} \cdot z_{i,t}$$

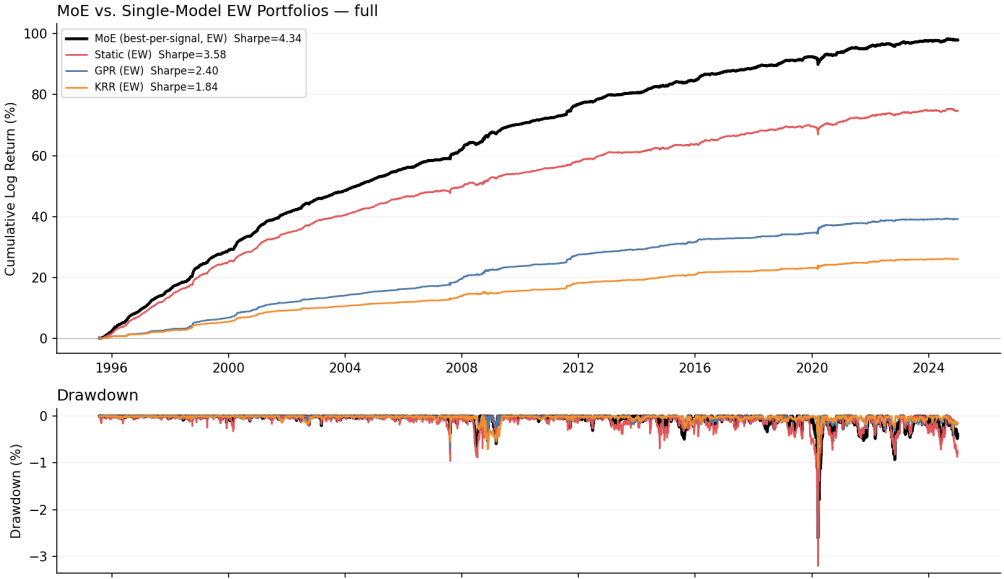
Optimization. Mean-variance optimization converts alphas into weights. Constraints: dollar-neutral (equal long/short) and market-neutral (zero net beta). Returns reflect pure signal skill with no directional market exposure.

Pipeline.

1. Each of the 8 signals runs through its own model independently
2. Each produces a daily long-short portfolio
3. Final portfolio = equal-weight average of all 8

Walk-forward: fit on past 504 days $\rightarrow$ predict $\rightarrow$ optimize $\rightarrow$ trade. No lookahead.

Results: Mixture of Experts (1995–2025)



Results: Portfolio Comparison

Mixture of Experts: for each signal, pick the model with the highest Sharpe, then equal-weight the 8 winners. GPR wins 6 of 8 signals. Static wins idiosyncratic reversal and volatility — signals with the most stable, linear IC.

Portfolio	**Cum. Return %**	**Ann. Return %**	**Sharpe**	**Max DD %**
MoE (best model per signal)	**97.9**	**3.33**	**4.34**	−2.61
Static (EW all signals)	74.7	2.54	3.58	−3.20
GPR (EW all signals)	39.1	1.33	2.40	−0.87
KRR (EW all signals)	26.0	0.89	1.85	−1.13

Ethical Implications

1. The Zero-Sum Nature of Alpha

Short-term alpha generation is essentially a zero-sum game. By executing trades based on predictive signals, the model systematically extracts wealth from less-informed market participants, raising questions about the broader societal value of these transactions.

2. Resource and Information Asymmetry

Deploying this strategy requires expensive proprietary data (e.g., Barra risk models) and significant daily compute. This structural barrier widens the wealth gap between institutional and retail investors, transferring returns to the better-equipped.

3. Societal Brain Drain

Quantitative finance aggressively recruits top-tier STEM talent by offering highly competitive compensation. This siphons intellectual capital away from research areas—such as healthcare, engineering, or climate science—that may offer higher tangible benefits to the public.

Conclusions

Central finding

$\mathrm{IC}_t(z)$ is non-constant. Kernel methods capture this — but the benefit is signal-specific.

- **KRR and GPR are nearly identical** in their predictions; GPR additionally provides posterior uncertainty used to size positions
- **Dynamic IC helps most** when IC sign is uncertain (factor-based reversal signals)
- **Static IC wins** when IC is stable and positive (idiosyncratic reversal)
- **Mixture of Experts** routes each signal to its best model: Sharpe $3.58 \to 4.34$

10 slides total